Utkarsh Sharma

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Research Interests: Differentiable & Neural Rendering, Physically-Based Rendering, Inverse Graphics

Education

Indian Institute of Science (IISc) — **Bangalore, India**
M.Tech in Computer Science and Engineering | ***CGPA: 9.1/10.0*** — *2024 – 2026*

Key Coursework: Graphics & Visualization, Learning for 3D Vision and Inverse Graphics, Deep Learning for Computer Vision, Computational Topology, Computational Methods of Optimization, Machine Learning for Signal Processing, Design and Analysis of Algorithms

Chandigarh University — **Mohali, India**
B.E. in Computer Science | ***CGPA: 8.24/10.0*** — *2020 – 2024*

Research Experience

M.Tech Thesis Project: Radiosity-Guided Accelerated Path Tracing | *CS 299, IISc* — **Jan – Apr 2026**

- Jointly advised by **Prof. Vijay Natarajan** (Visualization & Graphics Lab) and **Prof. R. Venkatesh Babu** (Vision & AI Lab).
- Proposed a camera-independent path guiding framework utilizing hardware-accelerated progressive radiosity to extract top-K radiant contributors to construct incoming radiance distribution.
- Designed a closed-form von Mises-Fisher (vMF) mixture model constructed on-the-fly, mapping precomputed radiometric geometry to continuous spherical distributions.
- Implemented a fused CUDA update kernel achieving bounded $\mathcal{O}(T)$ memory scaling and delivering up to 2.4× higher sample throughput than NPM and PPG baselines on Diffuse dominant scenes with comparable variance reduction.
- **Radiosity-Guided Accelerated Path Tracing – Utkarsh Sharma**, Prashant Goswami, Rishubh Parihar, Vijay Natarajan, R. Venkatesh Babu. *(Under review at JCGT, July 2026)*

Selected Research & Projects

Futaba: Hardware-Accelerated GPU Path Tracer | *C++17, CUDA, OptiX, OpenGL* — **Mar – Aug 2025**

- Developed a GPU-accelerated path tracer in C++/CUDA with NVIDIA OptiX hardware ray tracing and an optimized software BVH fallback, tuning thread occupancy and GPU memory layouts.
- Implemented Next Event Estimation with multiple importance sampling, Russian Roulette, thin-lens depth of field, environment-map importance sampling, and Beckmann microfacet BSDFs (conductor/dielectric/plastic).
- Built an interactive OpenGL viewport (zero-copy PBO display) with multi-channel G-buffer visualizations (albedo, normals, depth, heatmap) and an OptiX AI denoiser for debugging and final output quality.

Neural Path Guiding for Path Tracing | *Mitsuba 3, PyTorch, Tiny-CUDA-NN* — **Jun – Dec 2025**

- Built a modular neural path-guiding framework in Mitsuba 3 (Python), implementing vMF mixture models, Neural Importance Sampling, and Distribution Factorization to guide ray directions, validating 2–4× variance reduction in complex scenes.
- Developed a PyQt-based interactive GUI to inspect Mitsuba renders, visualize spherical probability density functions (PDFs), and debug neural network prediction behavior during rendering.
- Prototyped Neural Radiance Caching using online optimization and spatial hash-grid encodings.

Deep Learning-Based Denoising for Monte Carlo Rendering | *PyTorch, PBRT* — **Jan – Apr 2025**

- Evaluated and benchmarked diffusion-based and U-Net denoisers for low-sample Monte Carlo path tracing outputs in PyTorch and PBRT.
- Incorporated G-Buffer features (albedo, normals, depth, motion vectors) as multi-channel inputs to guide predictions, preserving geometric edges and material boundaries.
- Analyzed reconstruction quality using SSIM and PSNR metrics on PBRT-generated test scenes.

Neural Radiance Fields & 3D Gaussian Splatting | *PyTorch* — **Jan – May 2025**

- Built a NeRF pipeline from scratch in PyTorch, implementing the volumetric rendering equation, sinusoidal positional encoding, and hierarchical stratified sampling to reconstruct 3D scenes.
- Implemented 3D Gaussian Splatting, optimizing 3D covariance matrices and spherical harmonics (SH) coefficients; validated on the NeRF Synthetic dataset (PSNR $>$ 25 dB).

Technical Writing

Introduction to Differentiable Rendering **July 2026**

- Survey covering 6 seminal papers (Li 2018 through Zhang 2023), with step-by-step mathematical derivations, PyTorch implementations of Radiative and Path Replay Backpropagation.

Introduction to 3D Rendering **Apr 2026**

- Interactive graphics guide covering the rasterization pipeline, ray tracing and SDF sphere tracing implementations.

Importance Sampling & Path Guiding for Path Tracing **Dec 2025**

- Tutorial and survey spanning 9 papers (Vorba 2014 through Dahm 2025), covering Monte Carlo integration, MIS, normalizing flows, vMF mixtures, and neural path guiding.

3D Reconstruction with 3DGS, NeRF, and Their Variants **Nov 2025**

- Survey of 12 methods from NeRF to DUSt3R/InstantSplat, with derivations of volume rendering, EWA splatting, hash encoding, and sparse-view reconstruction.

Work Experience

Netradyne **Bangalore, India**

Machine Learning Engineer *July 2026 – Present*

- Analyze large-scale telematics and video-derived data to refine the **GreenZone® Score**, an AI-driven driver-performance metric, using Python, SQL, Pandas, and NumPy.

Skills & Languages

Programming Languages: Python, C, C++ (17/20), CUDA, SQL, GLSL
Graphics & 3D Vision: Physically Based Rendering (PBR), Monte Carlo Path Tracing, Differentiable Rendering, Neural Rendering (NeRF, 3DGS), Volume Rendering, NVIDIA OptiX, OpenGL
Machine Learning & AI: PyTorch, Diffusion Models, U-Net, Neural Radiance Caching
Libraries & Tools: Mitsuba 3, PBRT, Tiny-CUDA-NN, Dr.Jit, Pandas, NumPy, Git
Spoken Languages: Hindi (Native), English (Fluent)

Teaching Experience

Teaching Assistant – Algorithms & Programming | *Indian Institute of Science* **Aug – Dec 2025**
Teaching Assistant – Computational Topology | *Indian Institute of Science* **Jan – May 2026**

Academic Achievements

Rank 8 out of 123,967 candidates in GATE Computer Science (CS) 2024 *(India's national graduate admissions exam; top 0.01%)*
Rank 17 out of 39,210 candidates in GATE Data Science & AI (DA) 2024 *(top 0.05%)*
Winner, CodeSmash 2.0 Competitive Programming Contest
Finalist, CodeRush 3.0 Competitive Programming Contest